

Chap 1

1) Intermediary Network Devices

- An intermediary device interconnects end devices. Example, switches, routers, firewalls
- Management of data as it flows through a network is also the role of
 - Regenerate and retransmit data signals
 - ~~Maintain information about what pathways exist in the network~~
 - Notify other devices of errors and communication failures.

2) Internet, intranet, extranets

Internet

- The internet is a worldwide collection of interconnected LANs and WANs.

Intranet

- Intranet is a private network accessible only to an organization's internal members.

Extranet

- Extranet is a private network that allows authorized external users to access specific data.

Chap 2

Layered Model

// 7 OSI Reference Model →

1) Benefit of using a Layered Model

- Foster competition because products from different vendors can work together.
- Prevent technology or capability changes in one layer from affecting other layers above and below.

2) PDU

- Encapsulation is the process where protocols add their information to the data

PDUs passing down the stack are as follow:

- | | | |
|------------|-----------|---------|
| 1. Data | 3. Packet | 5. Bits |
| 2. Segment | 4. Frame | |

Chap 2 3) 7 OSI Reference Model layer

(continue)

OSI Model layer

Description

(7)	Application	Contains protocols used for process-to-process communications
(6)	Presentation	Provides for common representation of the data transferred between application layer services.
(5)	Session	Provides services to the presentation layer and to manage data exchange
(4)	Transport	Defines services to segment, transfer, and reassemble the data for individual communications.
(3)	Network	Provides services to exchange the individual pieces of data over the network
(2)	Data Link	Describe methods for exchanging data frames over a common media.
(1)	Physical	Describes the means to active, maintain, and de-active physical connections.

Chap 3 1) Bandwidth

— Bandwidth is the capacity at which a medium can carry data.

• Latency

— The time taken for data to travel from one point to another

• Throughput

— The measure of the transfer of bits across the media over a given period time.

• Goodput

— The measure of usable data transferred over a given period time.

— Goodput = Throughput - traffic overhead

2) Network Media (types of cable) / three types

— Unshielded Twisted-Pair (UTP) cable

— Shielded Twisted-Pair (STP) cable

— Coaxial Cable

Chap 3 (Continue) ~~20~~ ~~Net~~

2) Network Media (cont)

• Unshielded Twisted Pair (UTP)

- UTP is the most common networking media
- & terminated with RJ-45 connections

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Key characteristics of UTP

- Twisted pair protect the signal from interference

• Shielded Twisted Pair (STP)

- Better noise protection and more expensive than UTP
- & terminated with RJ-45 connections

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Key characteristics of STP

- The outer jacket protects the copper ~~wires~~ wires from physical damage.

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~~• Coaxial Cable - Key characteristics of coaxial cable~~~~The outer~~

3) Straight-through cables

- switch to router
- Hub to server
- switch to PC/server

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4) Crossover cables

- switch to switch
- Hub to hub
- Router to router
- PC to PC
- Switch to hub
- Router to PC

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Chap 4 Data Link Layer

The data link layer consists of two sublayers

1) Logical Link Control (LLC)

- The LLC sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers.

Media Access Control (MAC)

- The MAC sublayer is responsible for data encapsulation and media access control.

2) Physical and Logical Topologies

Physical topology

- show physical connections and how devices are interconnected.

Logical topology

- identifies the virtual connections between devices using device interfaces and IP addressing schemes.

3) Physical WAN Topologies

Point-to-point

- the simplest and most common WAN topology. Consists of a permanent link between two endpoints.

Hub and spoke

- similar to a star topology where a central site interconnects branch sites through point-to-point links.

Mesh

- provides high availability but requires every end system to be connected to every other end system

3) LAN ~~vs~~ vs WAN

- LANs have a high data transfer rate
- WANs have a lower data transfer rate compared to LANs

Chap 4 (cont.)

4) Contention-Based Access // Access control method

• CSMA/CD

— Used by legacy Ethernet LANs

— Operates in half-duplex mode where only one device sends or receives at a time

CSMA/CD collision detection process

— Devices detect the collision

• CSMA/CA

— Used by IEEE 802.11 WLANs

— Operates in half-duplex mode where only one device sends or receives at a time

CSMA/CA

— When transmitting, devices also include the time duration needed for the transmission.

Chap 5

1) ~~Ethernet~~ Frame MAC Address // switching Process Learn and Forward

• Learn / Examining the Source MAC address

— Switches examine all incoming frames for new source MAC address information to learn.

— If the source MAC address does exist, the switch updates the refresh timer for that entry

• Forward / Examining the Destination MAC Address

— If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port.

— If the destination MAC address is a unicast address, the switch will look for a match in its MAC address table.

Chap 5 (cont.)

2) ARP

ARP purpose

- To determine the destination MAC address, the device uses ARP.

ARP provides two basic functions

- Resolving IPv4 addresses to MAC addresses
- Maintaining an ARP table of IPv4 to MAC address mappings.

ARP functions

- Ethernet devices refer to an ARP table in its memory to find the MAC address that is mapped to the IPv4 address.

- A device will search its ARP table for a destination IPv4 address and a corresponding MAC address.

3) Video demonstration - ARP Request

- An ARP request is a broadcast frame sent when a device needs a MAC address associated with an IPv4 address, and it does not have an entry for the IPv4 address in its ARP table.

The ARP message include

- Target IPv4 address
- Target MAC address